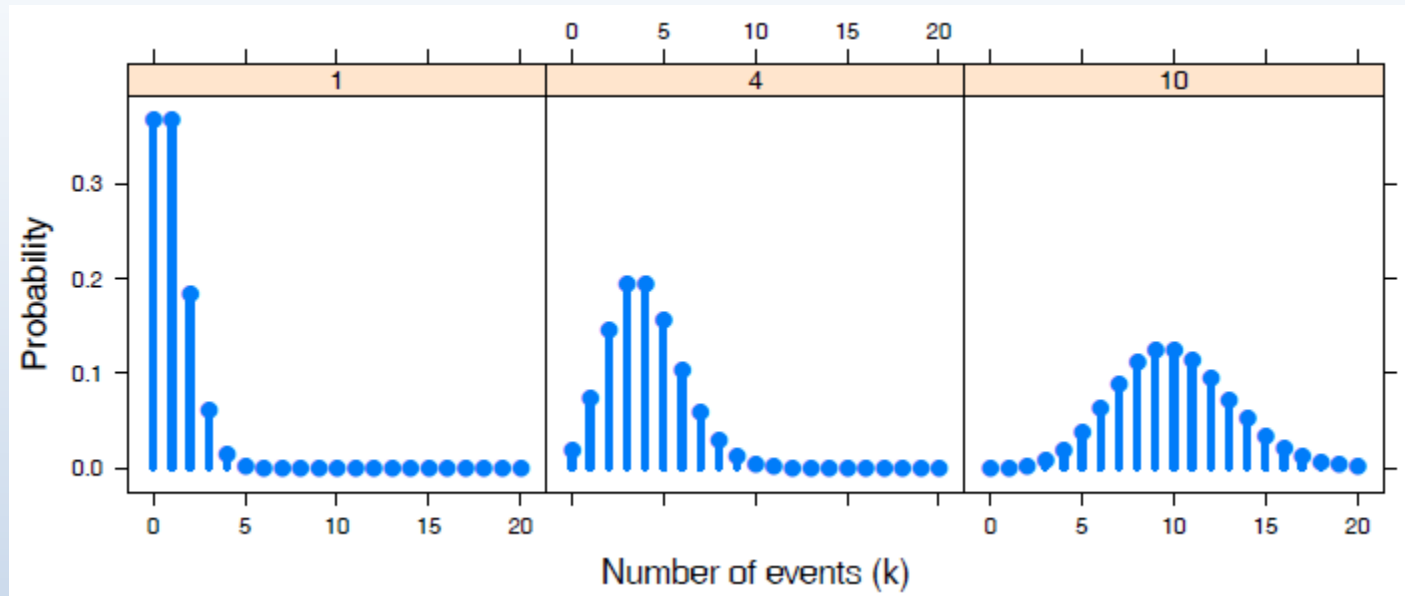
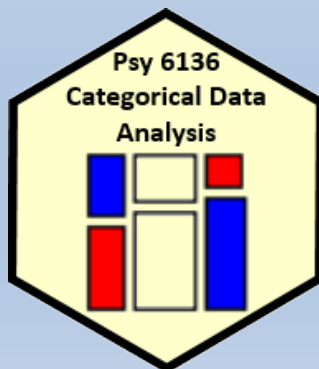




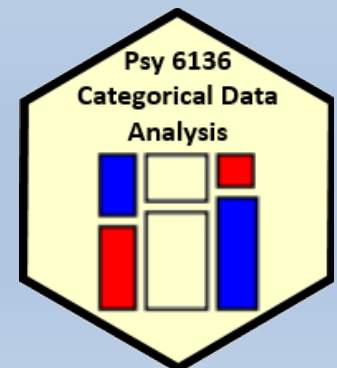
Discrete distributions



Michael Friendly

Psych 6136

<https://friendly.github.io/6135/>



Discrete distributions: Basic ideas

- Quantitative data: often assumed Normal (μ, σ^2) – **unreasonable** for CDA
- Binomial, Poisson, Negative binomial, ... are the building blocks for CDA
- They form the basis for modeling techniques
 - logistic regression, generalized linear models, Poisson regression
- Data: One-way tables
 - outcome variable ($k = 0, 1, 2, \dots$)
 - counts of occurrences (n_k): accidents, words in text, males in families of size k

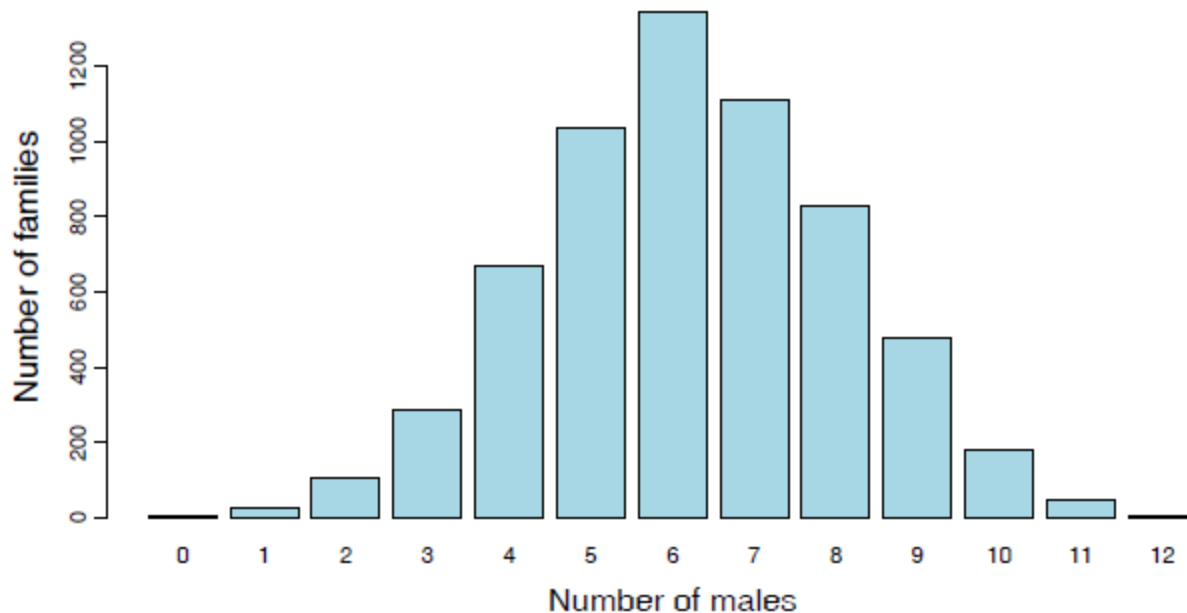
Examples: binomial?

Human sex ratio (Geissler, 1889): Is there evidence that $\Pr(\text{male}) = 0.5$?

Saxony families

Saxony families with 12 children having $k = 0, 1, \dots, 12$ sons.

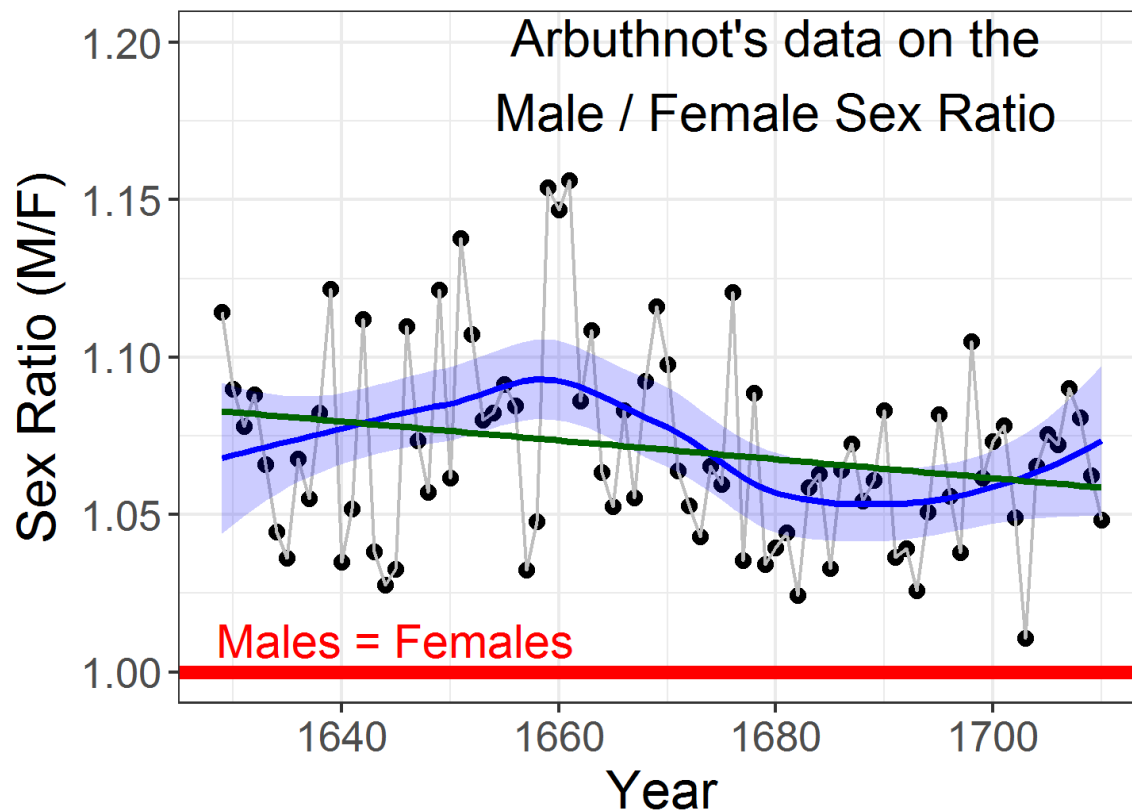
k	0	1	2	3	4	5	6	7	8	9	10	11	12
n_k	3	24	104	286	670	1033	1343	1112	829	478	181	45	7



Why did this matter? -- Arbuthot

John Arbuthot (1710) had gathered data on the ratios of male to female christenings in London from 1629-1710, 81 years

- He showed that in **EVERY** year Males > Females
- Calculated the FIRST significance test: like tossing 81 coins, and getting all heads
- Argued that the result must be due to divine providence: God likes more boys than girls!



Data: [HistData::Arbuthnot](#)

ggplot script:

<https://friendly.github.io/6135/R/arbutnot-gg.html>

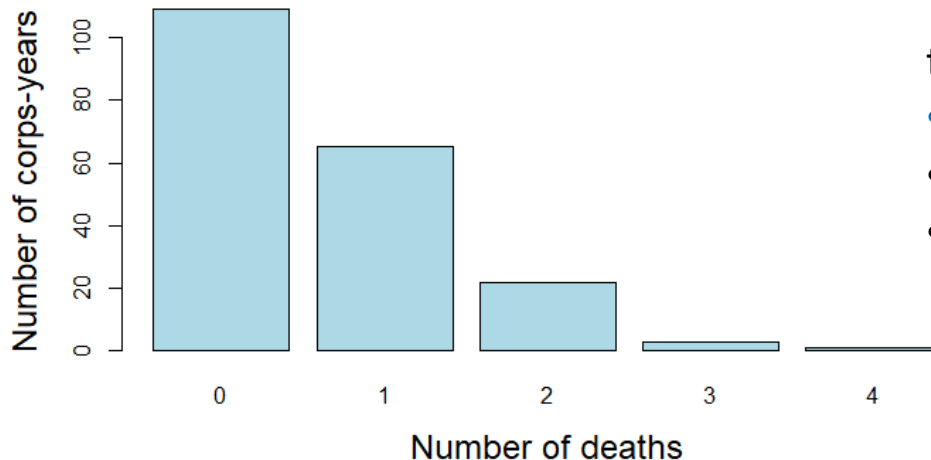
Example: Poisson

L. Von Bortkiewicz (1898) tallied the numbers of deaths by horse or mule kicks in 10 corps of the Prussian army over 20 years, → 200 corps-years

- In how many corps-years were there 0, 1, 2, ... deaths?
- This is among the earliest examples of a Poisson distribution



```
> data(HorseKicks, package="vcd")  
> HorseKicks  
nDeaths  
  0    1    2    3    4  
109   65   22    3    1
```



The Poisson distribution arises as that of the probability of 0, 1, 2, ...

- **Rare** events, that
- Occur with **constant** probability
- Forms the basis for `glm()` models with a count outcome.

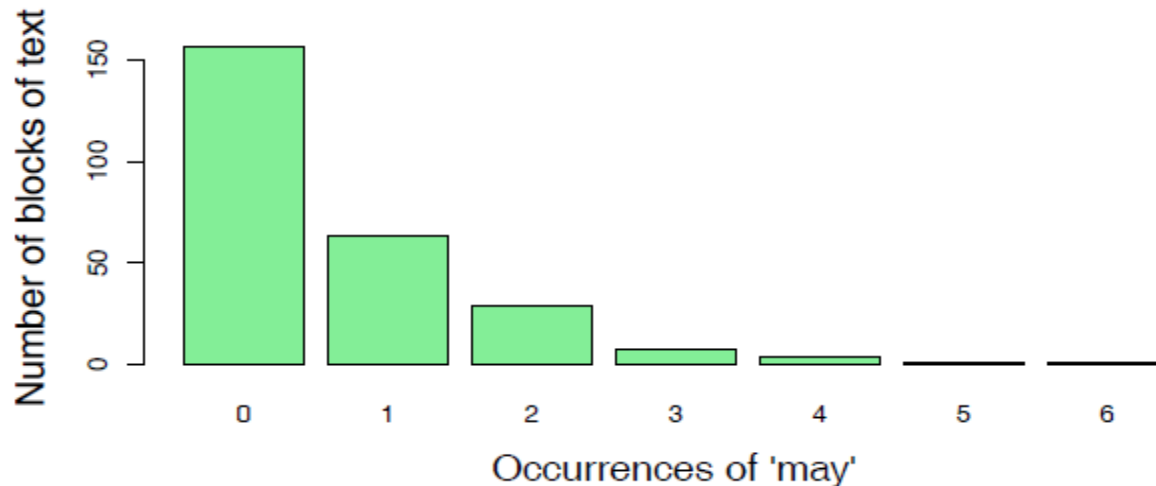
Examples: count data

Federalist papers: Disputed authorship

- 77 essays by Alexander Hamilton, John Jay, James Madison to persuade voters to ratify the US constitution, all signed with pseudonym “Publius”
 - Who wrote each?
 - 65 known, 12 disputed (H & M both claimed sole authorship)
- Mosteller & Wallace (1984): analysis of frequency distⁿs of key “marker” words: from, **may**, whilst, upon, ...
- e.g., blocks of 200 words: occurrences (k) of “may” in how many blocks (n_k)

```
> data(Federalist, package = "vcd")
> Federalist
nMay
  0    1    2    3    4    5    6
156  63  29   8   4   1   1
```

Count data: models



For each word (“from”, “may”, “whilst”, ...)

- Fit a probability model [$\text{Poisson}(\lambda)$, $\text{NegBin}(\lambda, p)$]
- Estimate parameters (λ, p)
- → Calculate log Odds (Hamilton vs. Madison)
- → All 12 disputed papers most likely written by **Madison**

(pioneered the use of cross-validation to assess model fit)

Example: Type-token distributions

- Basic count, k : number of “types”; frequency, n_k : number of instances observed
 - Frequencies of distinct words in a book or literary corpus
 - Number of subjects listing words as members of the semantic category “fruit”
 - Distinct species of animals caught in traps
- Differs from other distributions in that the frequency for $k = 0$ is *unobserved*
- Distribution is often extremely skewed (J-shaped)

Table: Number of butterfly species n_k for which k individuals were collected

Individuals (k)	1	2	3	4	5	6	7	8	9	10	11	12	
Species (n_k)	118	74	44	24	29	22	20	19	20	15	12	14	
Individuals (k)	13	14	15	16	17	18	19	20	21	22	23	24	Sum
Species (n_k)	6	12	6	9	9	6	10	10	11	5	3	3	5

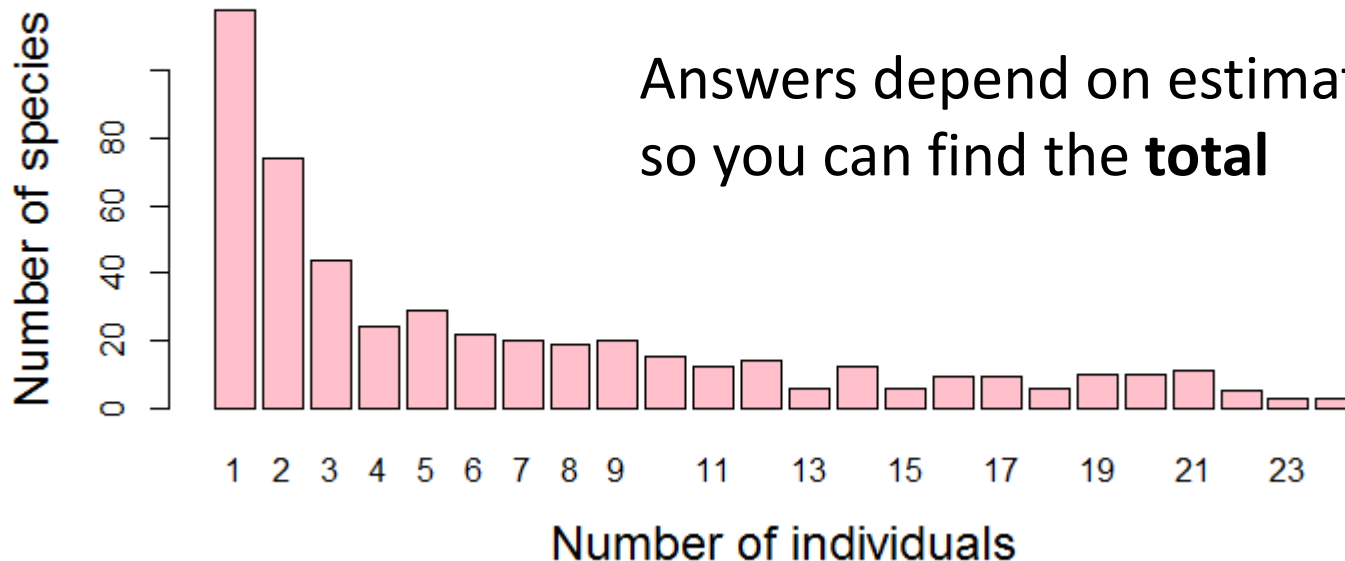

```
data(Butterfly, package="vcd")
barplot(Butterfly,
        xlab = "Number of individuals",
        ylab = "Number of species",
        col = "pink",
        cex.lab = 1.5)
```

Questions:

What is the total pop. of butterflies in Malaysia?

How many wolves remain in Canada NWT?

How many words did Shakespeare know?



Answers depend on estimating $\Pr(k=0)$,
so you can find the **total**

Discrete distributions: Questions

- General questions
 - What process gave rise to the distribution?
 - What is the form: uniform, binomial, Poisson, negative binomial, ... ?
 - → Fit & estimate parameters
 - Visualize goodness of fit
 - → Use in some larger context to tell a story
- Examples
 - *Families in Saxony*: might expect $\text{Bin}(n=12, p)$; $p=0.5$?
 - *HorseKicks*: Poisson (λ); here, $\lambda = \text{mean} = 0.61$
 - *Federalist papers*: Perhaps Poisson(λ) or NegBin (λ, p)
 - *Butterfly data*: Perhaps a log-series distribution?

Fitting discrete distributions

Lack of fit:

- Lack of fit tells us something about the *process* giving rise to the data
- Poisson: assumes constant small probability of the basic event
- Binomial: assumes constant probability and independent trials
- Negative binomial: allows for *overdispersion*, relative to Poisson

Motivation:

- Models for more complex categorical data use these basic discrete distributions
- Binomial (with predictors) → logistic regression
- Poisson (with predictors) → poisson regression, loglinear models
- ⇒ many of these are special cases of *generalized linear models*

Common discrete distributions

Distribution	Counts, k	Values of X	$\Pr(X=k)$	Mean, $E(X)$	Var, $V(X)$
Bernoulli(p)	Success in 1 trial	$k=\{0, 1\}$	$p^k(1-p)^{1-k}$	p	$p(1-p)$
Binomial(n, p)	# successes in n trials	$0, 1, \dots, n$	$\binom{n}{k} p^k(1-p)^{n-k}$	np	$np(1-p)$
Geometric(p)	# of trials to 1 st success	$0, 1, 2, \dots$	$p(1-p)^k$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$
Neg. binomial(k, p)	# of trials to k^{th} success	$0, 1, 2, \dots$	$\binom{n+k-1}{k} p^k(1-p)^n$	$\frac{k(1-p)}{p}$	$\frac{k(1-p)}{p^2}$
Poisson(λ)	# of events in interval	$0, 1, 2, \dots$	$\frac{\lambda^k e^{-\lambda}}{k!}$	λ	λ
Log series(p)	# of types observed	$0, 1, 2, \dots$	$\frac{p^k}{n \log(1-p)}$		

Discrete distributions: R

R functions: {d___, p___, q___, r___}

- d___ **density** function, $\Pr(X=k) = p(k)$
- p___ **cumulative probability**, $F(k) = \sum_{X \leq k} p(k)$
- q___ **quantile** function, find $k = F^{-1}(p)$, smallest value such that $F(k) \geq p$
- r___ **random** number generator

Discrete distribution	Density (pmf) function	Cumulative (CDF)	Quantile CDF^{-1}	Random # generator
Binomial	<code>dbinom()</code>	<code>pbinom()</code>	<code>qbinom()</code>	<code>rbinom()</code>
Poisson	<code>dpois()</code>	<code>ppois()</code>	<code>qpois()</code>	<code>rpois()</code>
Negative binomial	<code>dnbinom()</code>	<code>pnbinom()</code>	<code>qnbinom()</code>	<code>rnbinom()</code>
Geometric	<code>dgeom()</code>	<code>pgeom()</code>	<code>qgeom()</code>	<code>rgeom()</code>
Logarithmic series	<code>dlogseries()</code>	<code>plogseries()</code>	<code>qlogseries()</code>	<code>rlogseries()</code>

```
e.g., > dbinom(0:4, size=4, p=1/2)           # number of H in 4 coin tosses
[1] 0.0625 0.2500 0.3750 0.2500 0.0625
> dpois(0:4, lambda=3)                   # poisson, with  $\lambda = 3$ 
[1] 0.0498 0.1494 0.2240 0.2240 0.1680
```

What is “binomial”

Bi-no-mi-al /bī'nōmēəl/

- **Taxonomy:** A **two-part name**, (genus, species) e.g., *Elephas maximus* for the Asian elephant; *Canis familiaris* for domestic dog
- **Mathematics:** An algebraic expression of a sum of **two terms**, $(x + y)$ or expansion, $(x + y)^n$

$$\begin{aligned}(x+y)^0 &= 1 \\(x+y)^1 &= 1x + 1y \\(x+y)^2 &= 1x^2 + 2x^1y^1 + 1y^2 \\(x+y)^3 &= 1x^3 + 3x^2y^1 + 3x^1y^2 + 1y^3 \\(x+y)^4 &= 1x^4 + 4x^3y^1 + 6x^2y^2 + 4x^1y^3 + 1y^4 \\(x+y)^5 &= 1x^5 + 5x^4y^1 + 10x^3y^2 + 10x^2y^3 + 5x^1y^4 + 1y^5\end{aligned}$$

Coefficients of terms

$$C_k^n = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

(Pascal's triangle)

- **Stats:** A distribution of number of “successes” in n Bernoulli trials w/ prob = p

Binomial distribution

The binomial distribution, $\text{Bin}(n, p)$,

$$\text{Bin}(n, p) : \Pr\{X = k\} \equiv p(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad k = 0, 1, \dots, n, \quad (1)$$

Diagram annotations:

- Red arrow from $\binom{n}{k}$ to "# ways to get k out of n"
- Red arrow from p^k to "Pr(k events)"
- Red arrow from $(1-p)^{n-k}$ to "Pr(n-k non-events)"

arises as the distribution of the number of events of interest ("successes") which occur in n *independent trials* when the probability of the event on any one trial is the *constant* value $p = \Pr(\text{event})$.

Examples

- Toss 10 fair coins– how many heads? $\text{Bin}(10, \frac{1}{2})$
- Toss 12 fair dice– how many 5s or 6s? $\text{Bin}(12, 1/3)$

Mean, variance, skewness:

$$\text{Mean}[X] = n p$$

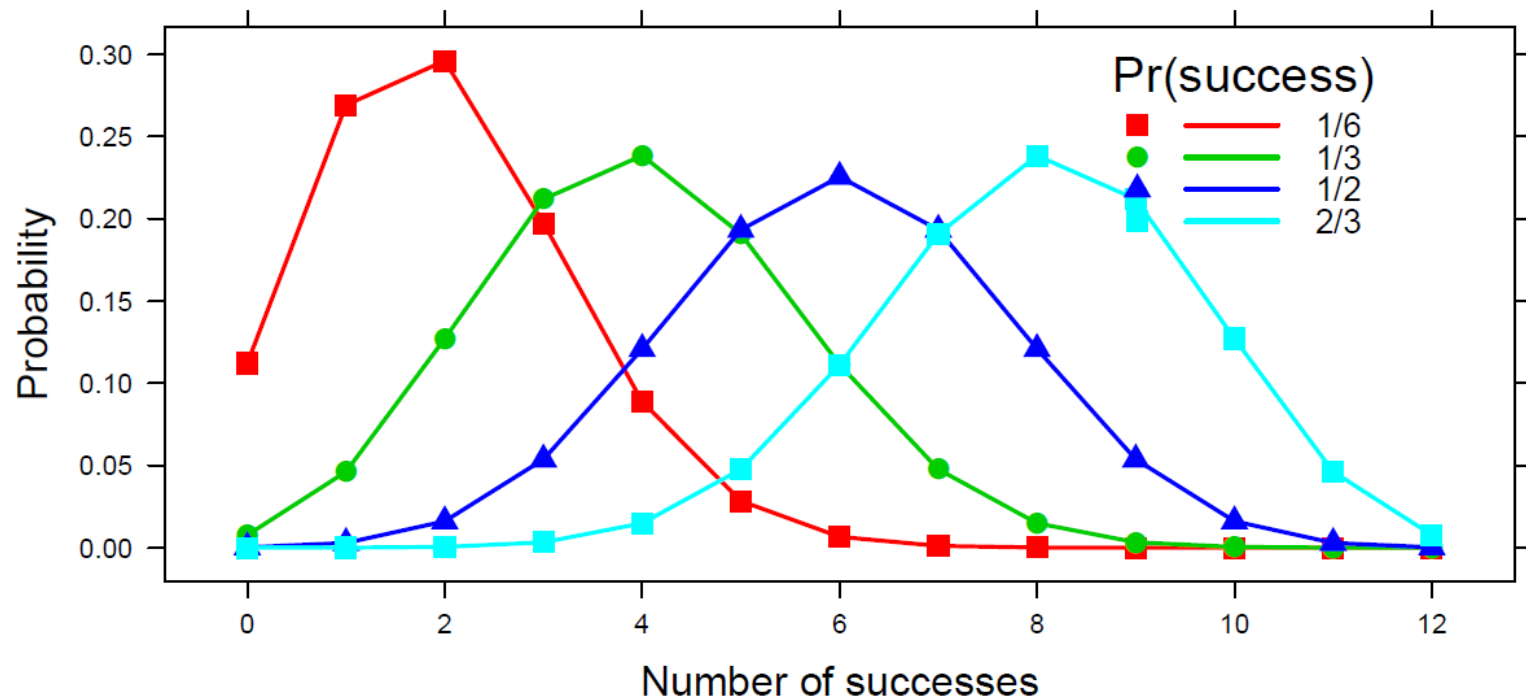
$$\text{Var}[X] = n p (1-p) = n p q$$

$$\text{Skew}[X] = n p q (q-p)$$

$$\text{MLE from data: } \hat{p} = \frac{\bar{x}}{n} = \frac{\sum_k k \times n_k / \sum_k n_k}{n}$$

Binomial distribution

Binomial distributions for $k = 0, 1, 2, \dots, 12$ successes in $n=12$ trials, for 4 values of p



- Mean = $n p$
- Variance: $n p q$ is maximum when $p = \frac{1}{2}$
- Skewed when $p \neq \frac{1}{2}$

Code: binomial.html

DDAR Fig 3.9, pp 76-77

Poisson distribution

The Poisson distribution, $\text{Pois}(\lambda)$,

$$\text{Pois}(\lambda) : \Pr\{X = k\} \equiv p(k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad k = 0, 1, \dots \quad (2)$$

gives the probability of an event occurring $k = 0, 1, 2, \dots$ times over a *large number of independent* trials, when the probability, p , that the event occurs on any one trial (in time or space) is *small and constant*.

Examples:

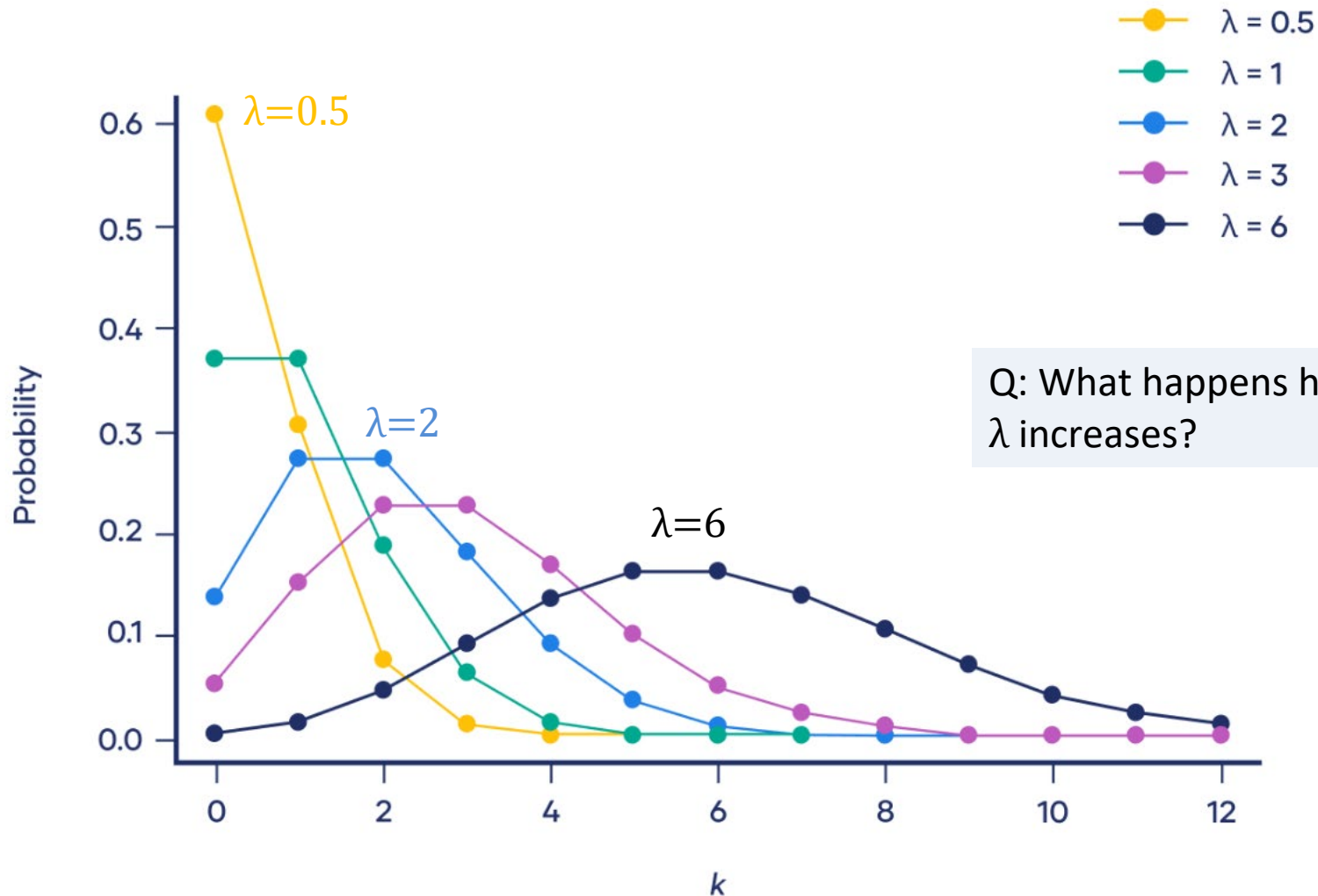
- Number of highway accidents at some given location
- Defects in a manufacturing process
- Number of goals scored in soccer games

Table: Total goals scored in 380 games in the Premier Football League, 1995/95 season

Total goals	0	1	2	3	4	5	6	7
Number of games	27	88	91	73	49	31	18	3

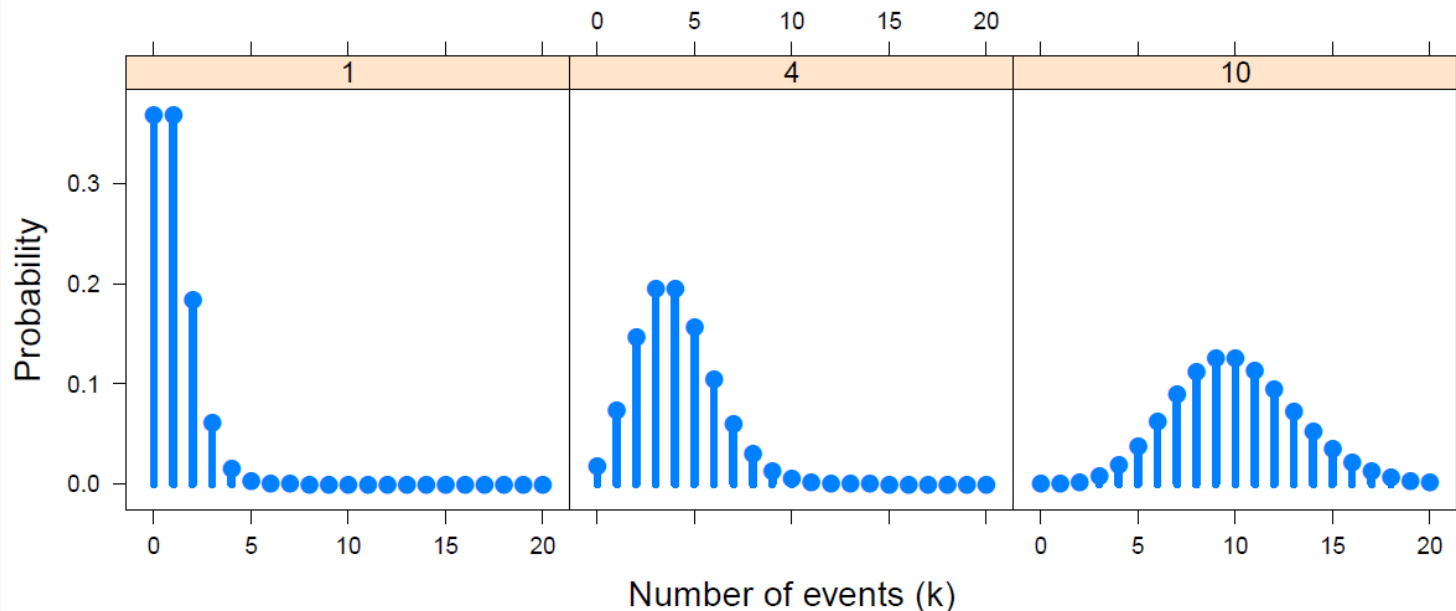
Poisson distribution

Poisson distributions for $\lambda = \frac{1}{2}, 1, 2, 3, 6$



Poisson distribution: Properties

Poisson distributions for $\lambda = 1, 4, 10$



DDAR Fig
3.10, p 81

Mean, variance, skewness:

$$\text{Mean}[X] = \lambda$$

$$\text{Var}[X] = \lambda$$

$$\text{Skew}[X] = \lambda^{-1/2}$$

$$\text{MLE: } \hat{\lambda} = \bar{x}$$

Properties:

Sum of Pois ($\lambda_1, \lambda_2, \lambda_3, \dots$) = Pois($\sum \lambda_i$)

Approaches $N(\lambda, \lambda)$ as $n \rightarrow \infty$

History: Who discovered the “Poisson” distribution

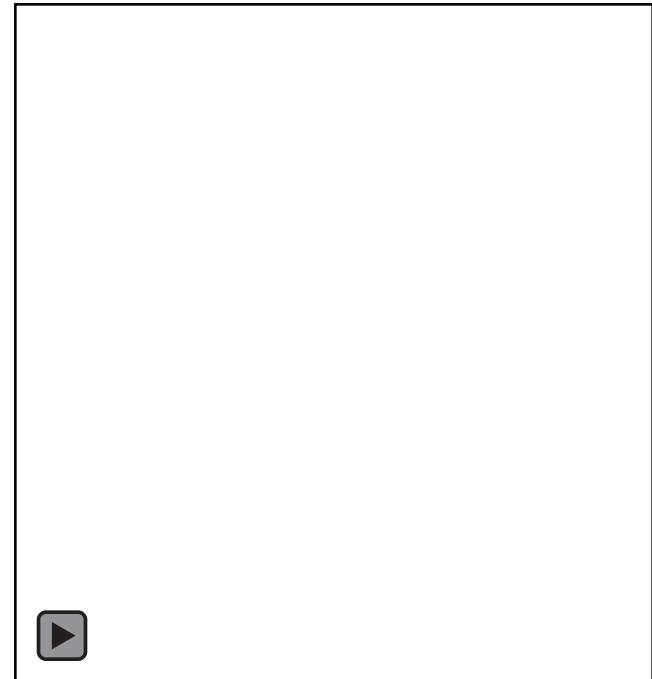
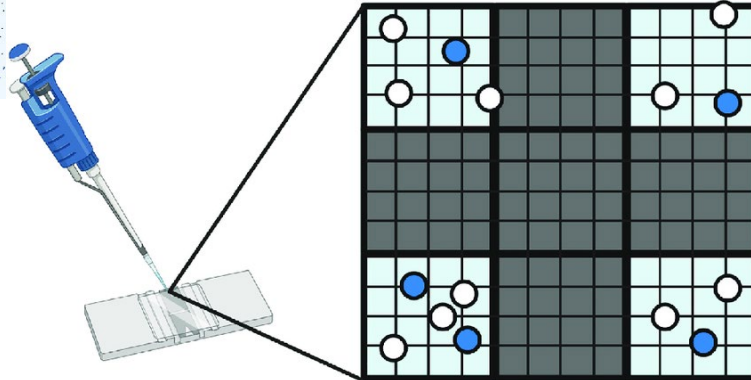
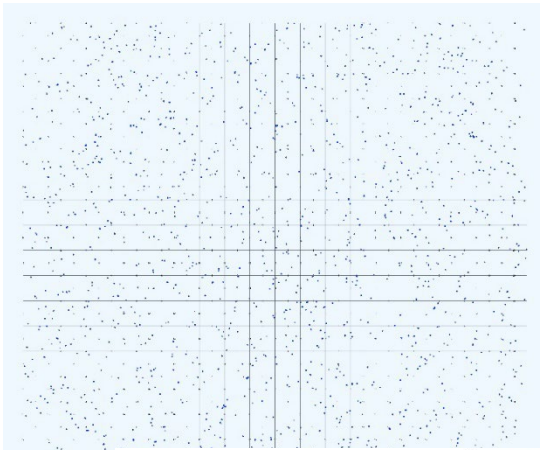
Stigler’s Law: No discovery in science is ever named for its primary originator

- De Moivre (1718) – Approximation to binomial as n gets largish
- **Poisson (1837)** – *Reserches sur la Probabilité des jugements en Matière criminelle...* -- Derives $e^{-\lambda} \lambda^k / k!$
 - Stigler says main result anticipated by De Moivre
- S. Newcomb (1860) – *Notes on a Theory of Probability*
 - First attempt at using this as a fit to **data**
 - Observations of stars: $\Pr(\text{any small space, } 1^\circ) \text{ contains } s \text{ stars, } s = 0, 1, 2, \dots$
- Von Bortkiewicz (1898) – *Law of Small Numbers*
 - Re-derives Poisson as limiting case of binomial
 - Several data sets (Horse kicks & others) – “agreement between theory and observation leave nothing to be desired”
- Gosset (1907): Heamacytometer Counts
 - “Student”’s first paper – first rigorous treatment of the Poisson for count data

See: Hanley & Bhatnagar (2022) The “Poisson” Distribution: History, Reenactments, Adaptations, *The American Statistician*, 76:4, 363-371, [DOI: 10.1080/00031305.2022.2046159](https://doi.org/10.1080/00031305.2022.2046159)

Gosset: Heamacytometer Counts

Number of blood cells observed in a 20 x 20 grid on a slide



Source: <http://www.medicine.mcgill.ca/epidemiology/hanley/Gosset/>

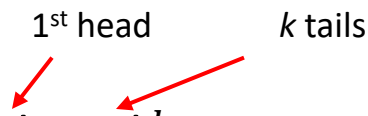
Geometric distribution

- Distribution of the **number of trials** **before** the **first occurrence** of some event
 - Flip a coin: how many trials before 1st head?
 - Toss a die: number of tosses before 1st “6”?
 - Quality control: number of items inspected before finding a defective one

Geom(p): $\Pr\{X=k\} = p(1-p)^k$

Mean: $E(X) = \frac{1-p}{p}$

Variance: $V(X) = \frac{1-p}{p^2}$



Assumptions:

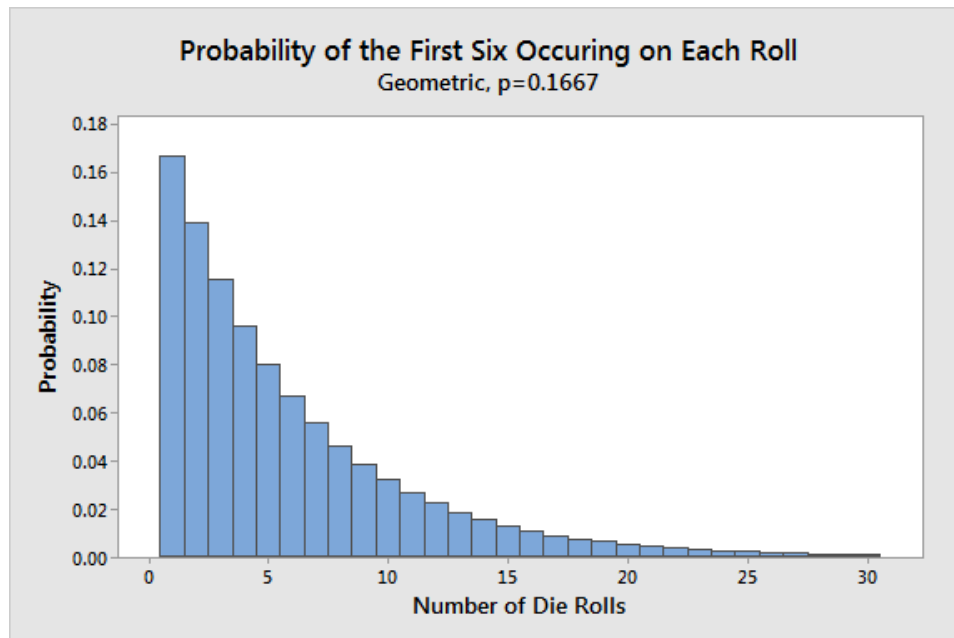
- Binary trials: Event / No event
- Independent trials
- Constant prob, p , on any trial

Geometric distribution: Tossing dice

```
> p <- 1/6
> x <- 0:10
> probs <- dgeom(x, prob = p)
> names(probs) <- x
> round(probs, 3)
```

0	1	2	3	4	5	6	7	8	9	10
0.167	0.139	0.116	0.096	0.080	0.067	0.056	0.047	0.039	0.032	0.027

For `dgeom()`, x is failures before first success, so $k=1$ (success on trial 1) means $x=0$ failures



Q: Can you use this to test if dice are “fair”, i.e., $p(k) = 1/6$?
How?

What dist^n would give an easier test?

Ans: Binomial– obs. counts of 1:6

Negative binomial distribution

The Negative binomial distribution, $\text{NBin}(n, p)$,

$$\text{NBin}(n, p) : \Pr\{X = k\} \equiv p(k) = \binom{n+k-1}{k} p^n (1-p)^k \quad k = 0, 1, \dots, \infty$$

$\Pr(n \text{ success})$ $\Pr(k \text{ failures})$

arises when a series of independent Bernoulli trials is observed with constant probability p of some event, and we ask how many non-events (failures), k , it takes to observe n successful events.

Example: Toss a coin; what is probability of getting $k = 0, 1, 2, \dots$ tails before $n = 3$ heads?

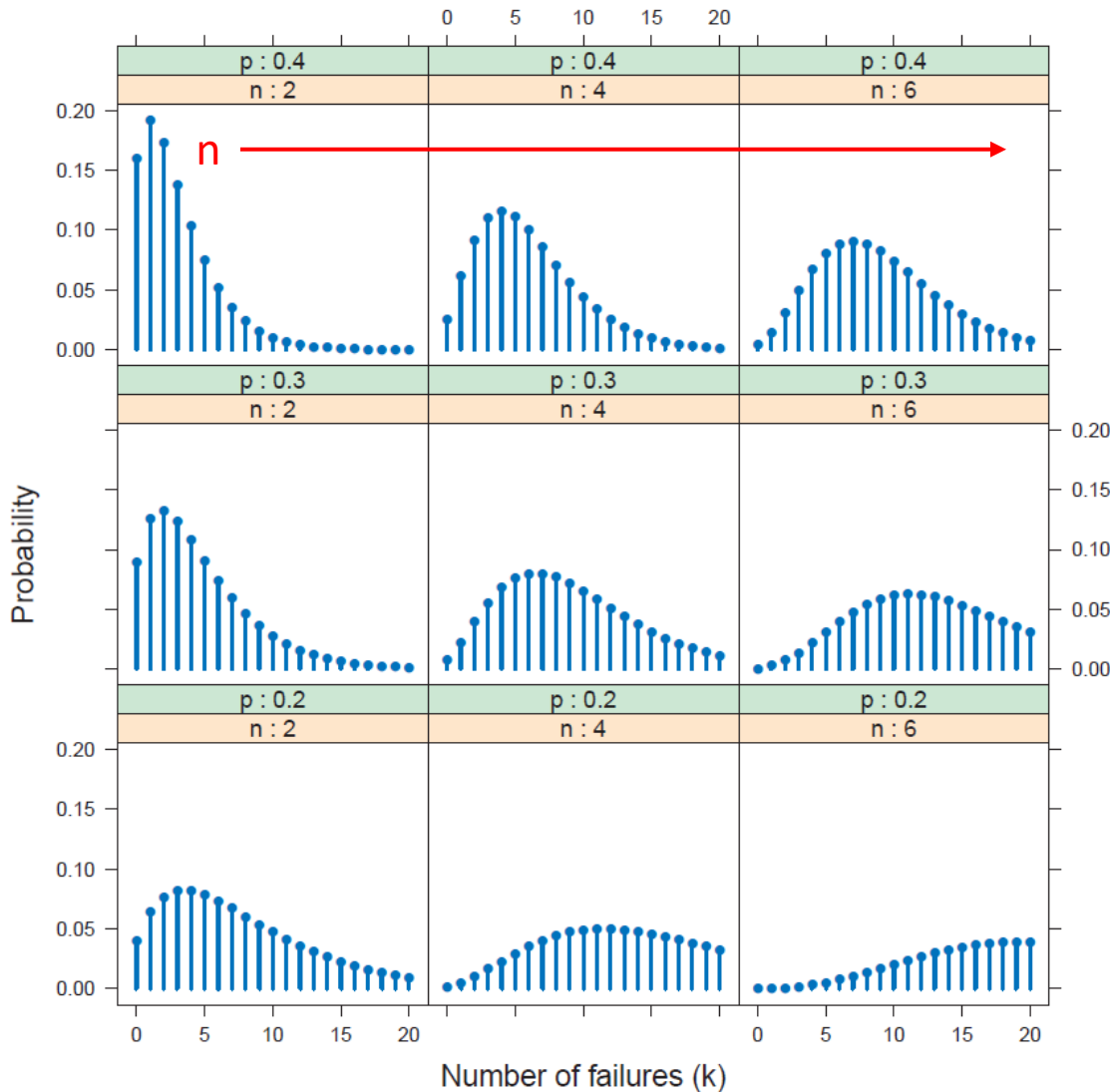
This distribution is often used as an alternative to the Poisson when

- constant probability p or independence are violated
- variance is greater than the mean (overdispersion)

Properties:

$$\text{Mean}(X) = \mu = \frac{n(1-p)}{p} \implies p = \frac{n}{n + \mu},$$

$$\text{Var}(X) = \frac{n(1-p)}{p^2} \implies \boxed{\text{Var}(X) = \mu + \frac{\mu^2}{n}}.$$



Negative binomial
distributions for
 $n = 2, 4, 6$
 $p = 0.2, 0.3, 0.4$

Mean:
Increases with n
Decreases with p

DDAR Fig 3.13, p 85

Quiz: Name that distribution

1. Weldon tossed 12 dice 26,306 times & tallied the number of times a **5 or 6** occurred

```
> WeldonDice
```

```
n56
```

0	1	2	3	4	5	6	7	8	9	10+
185	1149	3265	5475	6114	5194	3067	1331	403	105	18

Bin(n=12, p=1/3)

2. Pele practices penalty kicks for the upcoming 1958 FIFA World Cup. His average scoring has been $p=0.4$. What is the probability it will take him **1, 2, ... shots** to score a goal?

Nbin(n=1, p=0.4)

```
> dnbinom(1:5, size=1, p=0.4)
[1] 0.240 0.144 0.086 0.052 0.031
```

3. A Geiger counter records the number of scintillations of α particles from a radioactive source, with an average rate of 20/msec. What is the probability of observing 40 in a 1 msec. interval?

Pois($\lambda=20$)

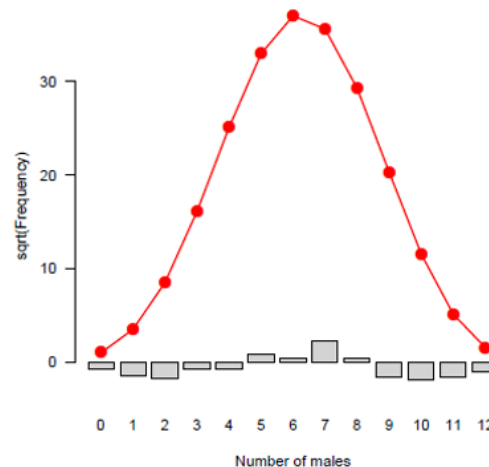
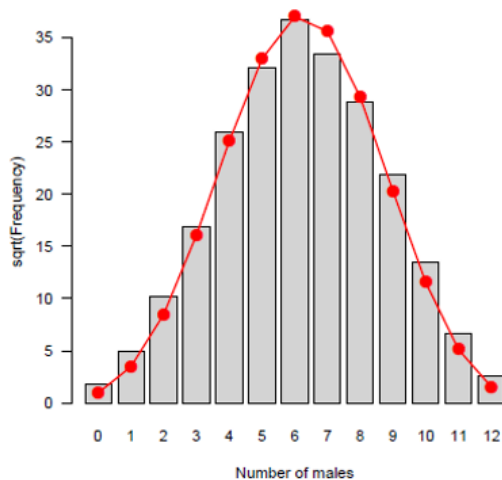
4. What is the distribution of the **time between** Geiger counter ticks?

Exponential distⁿ, $\Pr(X=k) = \lambda e^{-\lambda k}$, mean = $1/\lambda$

Fitting discrete distributions

Fitting discrete distributions generally follows four main steps:

- **Visualize the data** to suggest candidate distributions.
- **Estimate the parameters** of the chosen distribution.
- **Evaluate the quality of the fit** graphically.
- **Perform goodness-of-fit statistical tests** to determine how well the chosen model represents the data



Goodness-of-fit test for

	X^2	df	$P(> X^2)$
Likelihood Ratio	97.11	11	6.98e-16

Fitting discrete distributions: Details

Fitting a discrete distribution involves the following steps:

- 1 Estimate the parameter(s) from the data, e.g., p for binomial, λ for Poisson, etc. Typically done using maximum likelihood, but some distributions have simple expressions:
 - Binomial, $\hat{p} = \sum kn_k / (n \sum n_k) = \text{mean} / n$
 - Poisson, $\hat{\lambda} = \sum kn_k / \sum n_k = \text{mean}$
- 2 Calculate fitted probabilities, $\hat{p}(k)$ for the distribution, and then fitted frequencies, $N\hat{p}(k)$.
- 3 Assess Goodness of fit: Pearson X^2 or likelihood-ratio G^2

$$X^2 = \sum_{k=1}^K \frac{(n_k - N\hat{p}_k)^2}{N\hat{p}_k} \quad G^2 = \sum_{k=1}^K n_k \log\left(\frac{n_k}{N\hat{p}_k}\right)$$

Both have asymptotic chisquare distributions, χ^2_{K-s} with s estimated parameters, under the hypothesis that the data follows the chosen distribution.

Fitting: Weldon's dice

Basic, naïve calculation of expected frequencies for a binomial distⁿ using `dbinom()`

```
> data(WeldonDice, package="vcd")
> Weldon.df <- as.data.frame(WeldonDice) # convert to data frame

> Prob <- dbinom(0:12, 12, 1/3)           # binomial probabilities
> Prob <- c(Prob[1:10], sum(Prob[11:13])) # sum values for 10+
> Exp= round(sum(WeldonDice)*Prob)        # expected frequencies
> Diff = Weldon.df[, "Freq"] - Exp        # raw residuals
> Chisq = Diff^2 / Exp                    # contribution to chisquare
> data.frame(Weldon.df, Prob=round(Prob,5), Exp, Diff, Chisq)
```

	n56	Freq	Prob	Exp	Diff	Chisq
1	0	185	0.00771	203	-18	1.596
2	1	1149	0.04624	1216	-67	3.692
3	2	3265	0.12717	3345	-80	1.913
4	3	5475	0.21195	5576	-101	1.829
5	4	6114	0.23845	6273	-159	4.030
6	5	5194	0.19076	5018	176	6.173
7	6	3067	0.11127	2927	140	6.696
8	7	1331	0.04769	1255	76	4.602
9	8	403	0.01490	392	11	0.309
10	9	105	0.00331	87	18	3.724
11	10+	18	0.00054	14	4	1.143

Doesn't calculate the MLE, $\hat{p}$
Manually sum $k \geq 10$

Fitting & graphing discrete distributions

In R, the **vcd** and **vcdExtra** packages provide functions to fit, visualize and diagnose discrete distributions

- **Fitting:** `goodfit()` fits uniform, binomial, Poisson, neg bin, geometric, logseries, ...
- **Graphing:** `rootogram()` assess departure between observed, fitted counts
- **Ord plot:** `Ordplot()` diagnose form of a discrete distribution
- **Robust plots:** `distplot()` handle problems with discrepant counts

Example: Saxony families

```
> data(Saxony, package="vcd")
```

```
> Saxony
```

```
nMales
```

0	1	2	3	4	5	6	7	8	9	10	11	12
3	24	104	286	670	1033	1343	1112	829	478	181	45	7

Use **goodfit()** to fit the binomial; test with **summary()**

```
> Sax.fit <- goodfit(Saxony, type = "binomial", par=list(size=12))
```

```
> summary(Sax.fit)
```

Goodness-of-fit test for binomial distribution

	X^2	df	P(> X^2)
Likelihood Ratio	97	11	6.98e-16

Doesn't
fit. Why?

Specify parameters

Example: Saxony families

The `print()` method for **goodfit** objects shows the details

```
> Sax.fit      # print
```

Observed and fitted values for binomial distribution
with parameters estimated by 'ML'

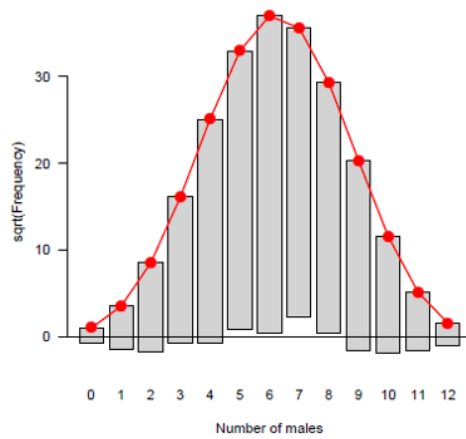
count	observed	fitted	pearson residual
0	3	0.933	2.140
1	24	12.089	3.426
2	104	71.803	3.800
3	286	258.475	1.712
4	670	628.055	1.674
5	1033	1085.211	-1.585
6	1343	1367.279	-0.657
7	1112	1265.630	-4.318
8	829	854.247	-0.864
9	478	410.013	3.358
10	181	132.836	4.179
11	45	26.082	3.704
12	7	2.347	3.037

Pay attention to the
signs & magnitudes of
residuals, d_k

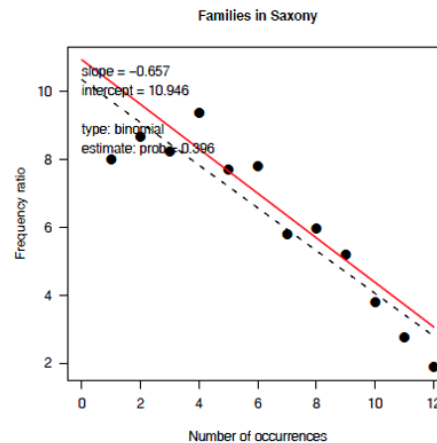
$$\text{Pearson } \chi^2 = \sum d_k^2$$

Graphing discrete distributions

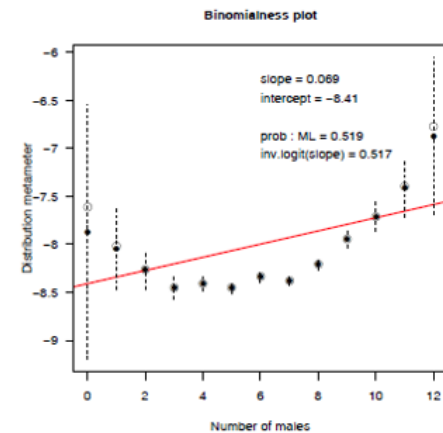
Rootograms



Ord plots



Robust
distribution
plots

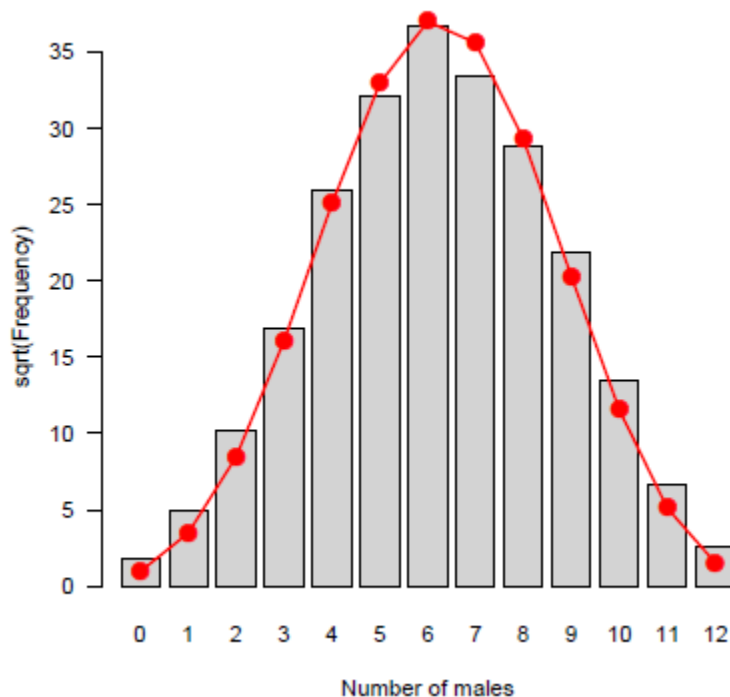


What's wrong with simple histograms?

Discrete distributions are often graphed as histograms, with a theoretical fitted distribution superimposed

The `plot()` method for `goodfit` objects provides some alternatives

```
> plot(Sax.fit, type = "standing", xlab = "Number of males")
```

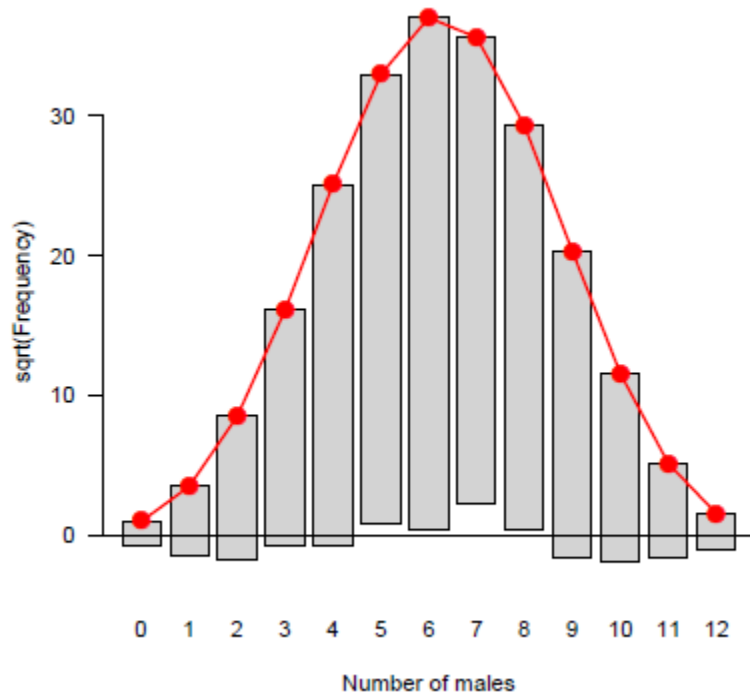


Problems:

- Largest frequencies dominate
- Must assess deviations vs. the fitted curve

Hanging rootograms

```
> plot(Sax.fit, type = "hanging", xlab = "Number of males") # default
```



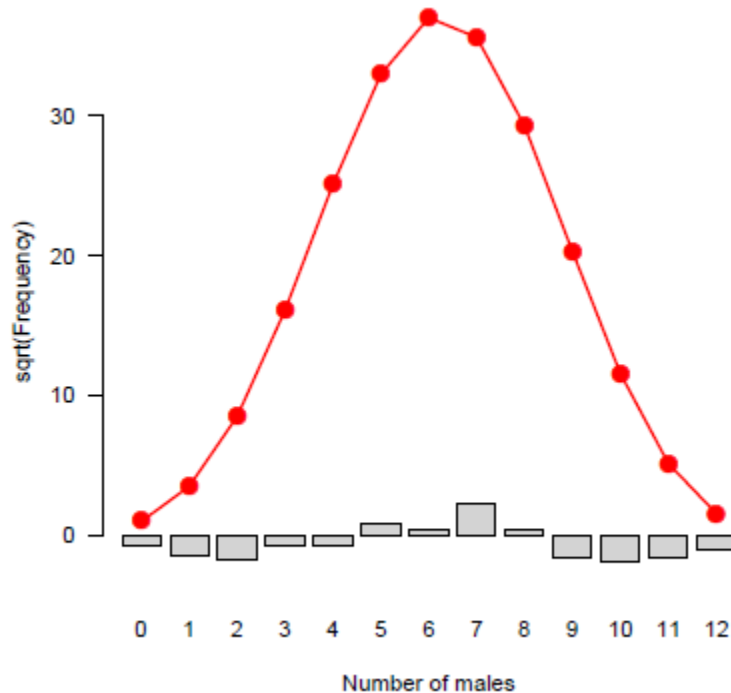
Tukey (1972, 1977):

- shift histogram bars to the fitted curve
- → judge deviations vs. horizontal line.
- plot $\sqrt{\text{freq}}$ → smaller frequencies are emphasized.

We can now see clearly *where* the binomial doesn't fit

Deviation rootograms

```
> plot(Sax.fit, type = "deviation", xlab = "Number of males")
```



Deviation rootogram:

- emphasize differences between observed and fitted frequencies
- bars now show the residuals (gaps) directly

There are more families with very low or very high number of sons than the binomial predicts.

Q: Why is this so much better than the lack-of-fit test?

Q: What does this suggest about sex distribution of families in Saxony?

Example: Federalist papers

```
> data(Federalist, package="vcd")  
> Federalist
```

nMay

0	1	2	3	4	5	6
156	63	29	8	4	1	1

Fit the Poisson distribution

```
> Fed.fit0 <- goodfit(Federalist, type="poisson")  
> summary(Fed.fit0)
```

Goodness-of-fit test for poisson distribution

	X^2	df	P(> X^2)
Likelihood Ratio	25.2	5	0.000125

This fits very poorly!
Why might this be the case?

Example: Federalist papers

Try the Negative binomial distribution

```
> Fed.fit1<- goodfit(Federalist, type="nbinomial")  
> summary(Fed.fit1)
```

Goodness-of-fit test for nbinomial distribution

	X^2	df	P(> X^2)
Likelihood Ratio	1.96	4	0.742

This now fits very well, indeed! Why?

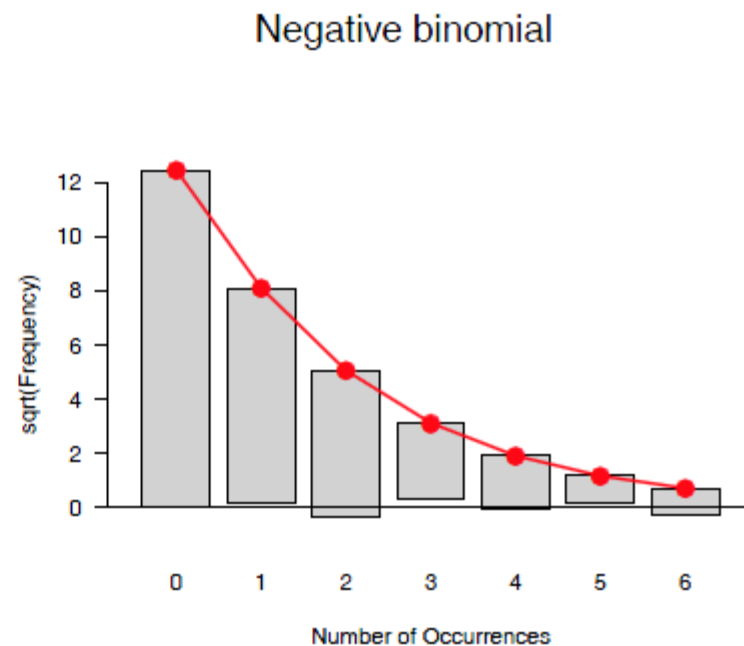
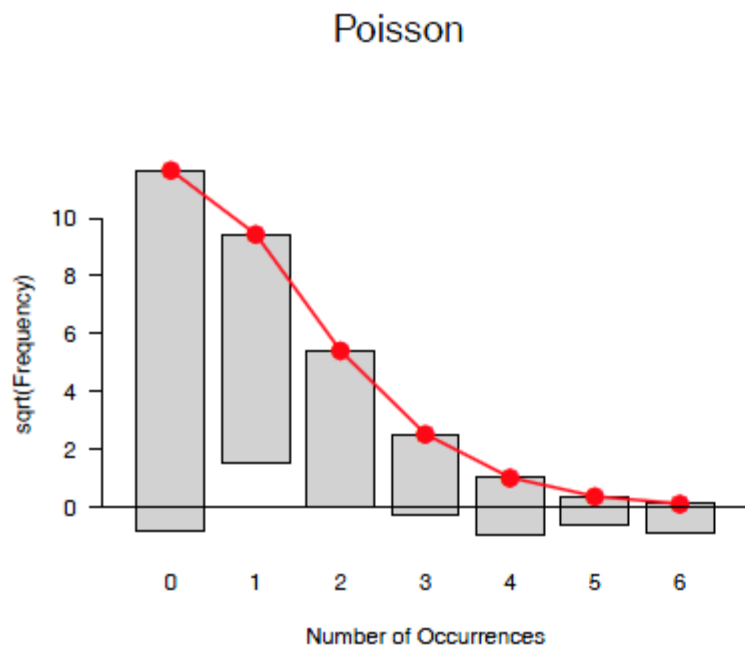
- Poisson assumes that the probability of a given word (“may”) is constant across all blocks of text.
- Negative binomial allows the rate parameter λ to vary over blocks of text

In their landmark study, Mosteller & Wallace used a number of such words to distinguish between papers written by Madison & Hamilton and then assign the “unknown” author papers --- all to James Madison

Federalist papers: Rootograms

Hanging rootograms for the Federalist papers data, comparing Poisson and Negative binomial

```
> plot(Fed.fit0, main = "Poisson")  
> plot(Fed.fit1, main = "Negative binomial")
```



History corner: Why important?

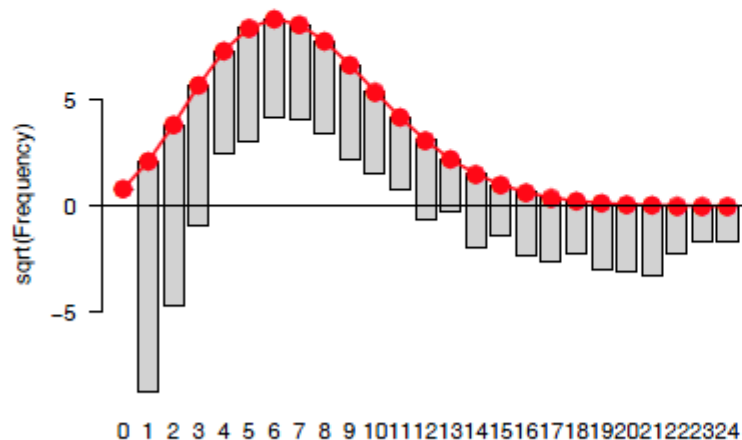
- Resolved a long-standing historical debate
 - Authorship of 12 of 85 papers had been disputed: Madison or Hamilton?
 - Analysis of word frequencies → strong evidence all by Madison
- Established “stylometry” – statistical analysis of text for authorship
 - Word frequencies of “marker” words (by, upon, may) as reliable indicators
 - Did Shakespeare write all his plays? Or did Francis Bacon or Christopher Marlow write some?
- Pioneered Bayesian analysis: First large-scale Bayesian application
 - Prior prob: $p(\text{Hamilton}) = p(\text{Madison}) = \frac{1}{2}$
 - Update via parameters estimated from $\text{Nbin}()$ distribution
- Introduced the idea of cross-validation
 - Leave-one-out → predict from the rest
- Heralded the age of computer-intensive methods

Butterfly data

Both Poisson and Negative binomial are terrible fits! What to do??

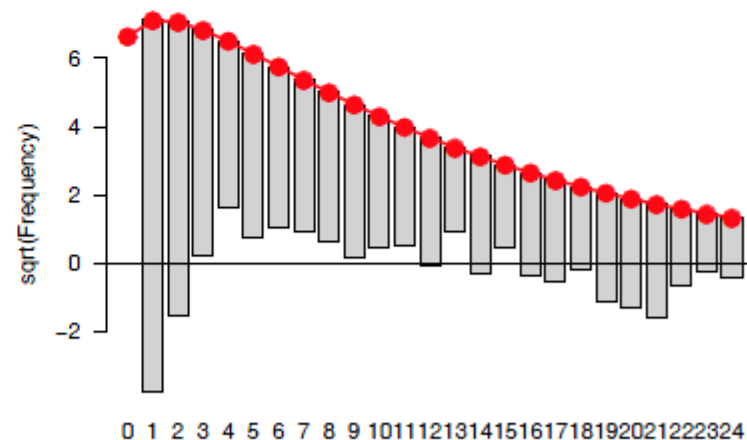
```
But.fit1 <- goodfit(Butterfly, type="poisson")  
But.fit2 <- goodfit(Butterfly, type="nbinomial")  
plot(But.fit1, main="Poisson")  
plot(But.fit2, main="Negative binomial")
```

Poisson



Number of Occurrences

Negative binomial



Number of Occurrences

Ord plots: Diagnose form of distribution

How to tell which discrete distributions are likely candidates?

- Ord (1967): for each of Poisson, Binomial, Negative binomial, and Logarithmic series distributions,

- plot of kp_k/p_{k-1} against k is linear
- signs of intercept and slope $\rightarrow$ determine the form, give rough estimates of parameters

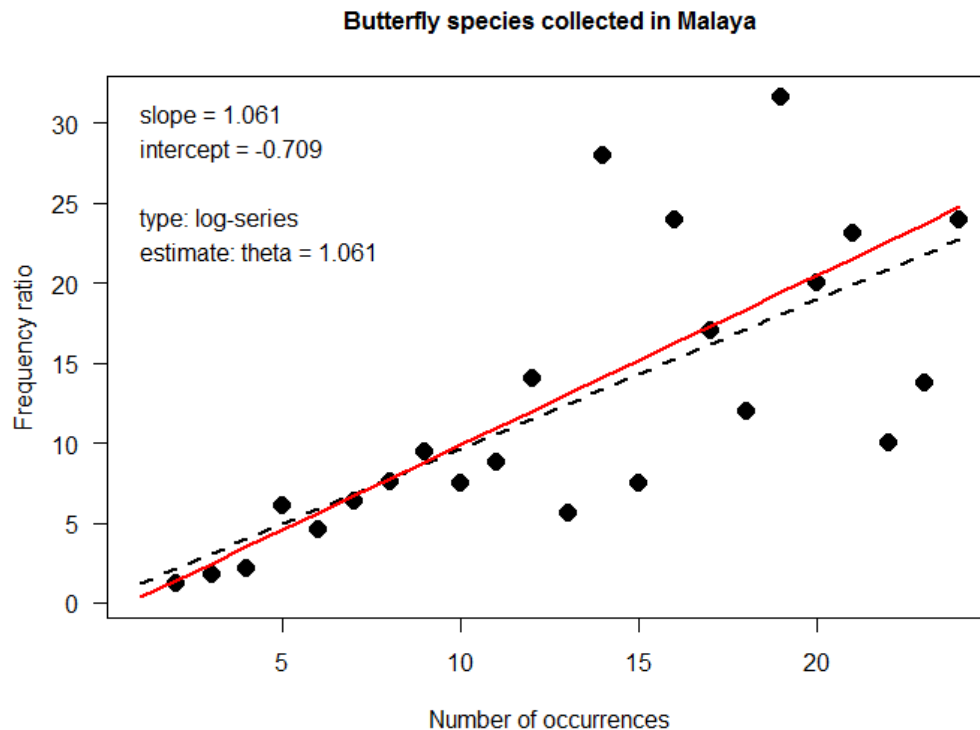
Slope (b)	Intercept (a)	Distribution (parameter)	Parameter estimate
0	+	Poisson (λ)	$\lambda = a$
—	+	Binomial (n, p)	$p = b/(b - 1)$
+	+	Neg. binomial (n, p)	$p = 1 - b$
+	—	Log. series (θ)	$\theta = b$ $\theta = -a$

- Fit line by WLS, using $\sqrt{n_k - 1}$ as weights
- A heuristic method: doesn't always work, but often a good start.

Ord plot: Examples

Butterfly data: The slope and intercept correctly diagnoses the **log-series** distribution

```
> Ord_plot(Butterfly,  
            main = "Butterfly species collected in Malaya",  
            gp=gpar(cex=1), pch=16)
```



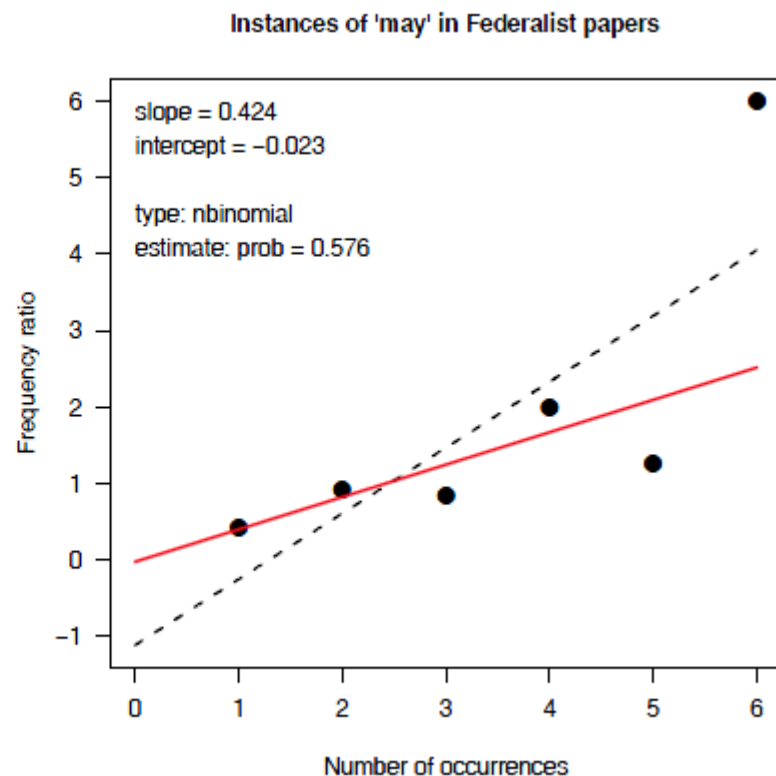
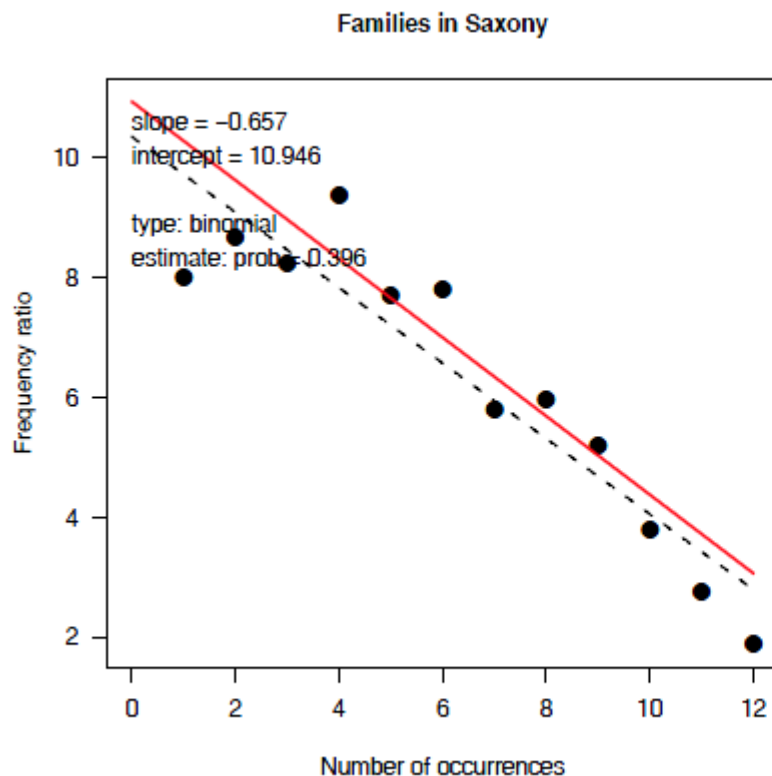
+ slope
- intercept
→ **log-series**

OLS line shown in black
WLS line shown in red

Ord plots: Examples

Ord plots for the Saxony and Federalist data

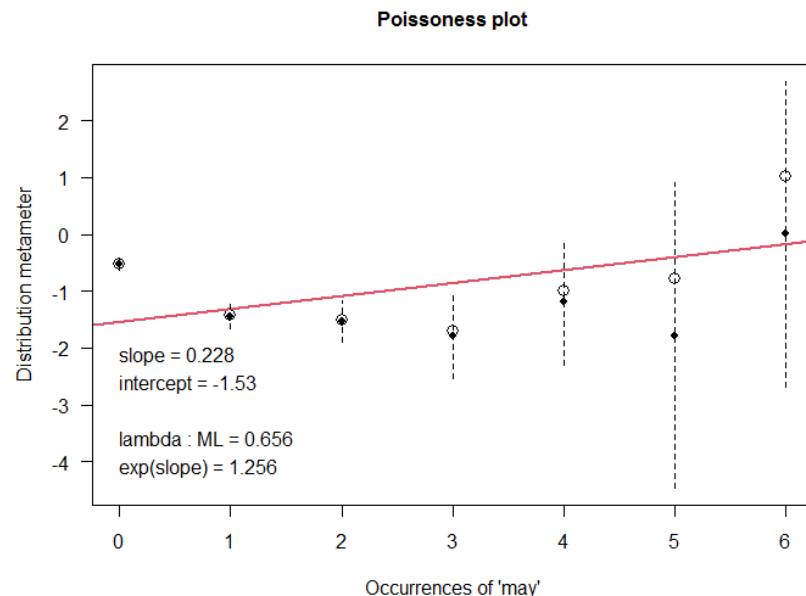
```
> Ord_plot(Saxony, main = "Families in Saxony", gp=gpar(cex=1), pch=16)  
> Ord_plot(Federalist, main = "Instances of 'may' in Federalist papers", gp=gpar(cex=1), pch=16)
```



Robust distribution plots

- Ord plots lack robustness
 - one discrepant frequency, n_k affects points for both k and $k + 1$
 - the use of WLS to fit the line is a small attempt to minimize this
- Robust plots for Poisson distribution (Hoaglin and Tukey, 1985)
 - For Poisson, plot **count metameter** $= \phi(n_k) = \log_e(k! n_k/N)$ vs. k
 - Linear relation $\Rightarrow$ Poisson, slope gives $\hat{\lambda}$
 - CI for points, diagnostic (influence) plot
 - Implemented in `distplot()` in the `ved` package

For the Poisson distribution, this is called a “poissonness plot”



Poissonness plot: Details

- If the distribution of n_k is $\text{Poisson}(\lambda)$ for some fixed λ , then each observed frequency, $n_k \approx m_k = Np_k$.
- Then, setting $n_k = Np_k = e^{-\lambda} \lambda^k / k!$, and taking logs of both sides gives

$$\log(n_k) = \log N - \lambda + k \log \lambda - \log k!$$

which can be rearranged to

$$\phi(n_k) \equiv \log \left(\frac{k! n_k}{N} \right) = -\lambda + (\log \lambda) k$$

- $\Rightarrow$ if the distribution is Poisson, plotting $\phi(n_k)$ vs. k should give a line with
 - intercept = $-\lambda$
 - slope = $\log \lambda$
- Nonlinear relation $\rightarrow$ distribution is *not* Poisson
- Hoaglin and Tukey (1985) give details on calculation of confidence intervals and influence measures.

Other distributions

This idea extends readily to other discrete data distributions:

- The binomial, Poisson, negative binomial, geometric and logseries distributions are all members of a general **power series family** of discrete distributions. See: *DDAR*, Table 3.10 for details.

$$p(k) = a(k) \theta^k / f(\theta) \quad k = 0, 1, \dots$$

Table 3.10: The Power Series family of discrete distributions

Discrete Distribution	Probability function, $p(k)$	Series parameter, θ	Series function, $f(\theta)$	Series coefficient, $a(k)$
Poisson	$e^{-\lambda} \lambda^k / k!$	$\theta = \lambda$	e^θ	$1/k!$
Binomial	$\binom{n}{k} p^k (1-p)^{n-k}$	$\theta = p/(1-p)$	$(1+\theta)^n$	$\binom{n}{k}$
Negative binomial	$\binom{n+k-1}{k} p^n (1-p)^k$	$\theta = (1-p)$	$(1-\theta)^{-k}$	$\binom{n+k-1}{k}$
Geometric	$p(1-p)^k$	$\theta = (1-p)$	$(1-\theta)^{-k}$	1
Logarithmic series	$\theta^k / [-k \log(1-\theta)]$	$\theta = \theta$	$-\log(1-\theta)$	$1/k$

Math trick: This allows properties of the members of a family to be studied, related to each other and solve larger problems!

Other distributions

- This allows all of these to be represented in a plot of a suitable count metameter, $\phi(n_k)$ vs k . DDAR Table 3.12 gives details

Distribution	Probability function, $p(k)$	Count metameter, $\phi(n_k^*)$	Theoretical slope (b)	Theoretical intercept (a)
Poisson	$e^{-\lambda} \lambda^k / k!$	$\log(k! n_k^* / N)$	$\log(\lambda)$	$-\lambda$
Binomial	$\binom{n}{k} p^k (1-p)^{n-k}$	$\log(n_k^* / N \binom{n}{k})$	$\log\left(\frac{p}{1-p}\right)$	$n \log(1-p)$
Negative binomial	$\binom{n+k-1}{k} p^k (1-p)^n$	$\log\left(n_k^* / N \binom{n+k-1}{k}\right)$	$\log(1-p)$	$n \log(p)$
Geometric	$p(1-p)^k$	$\log(n_k^* / N)$	$\log(1-p)$	$\log(p)$
Log series	$\theta^k / [-k \log(1-\theta)]$	$\log(k n_k^* / N)$	$\log(\theta)$	$-\log(-\log(1-\theta))$

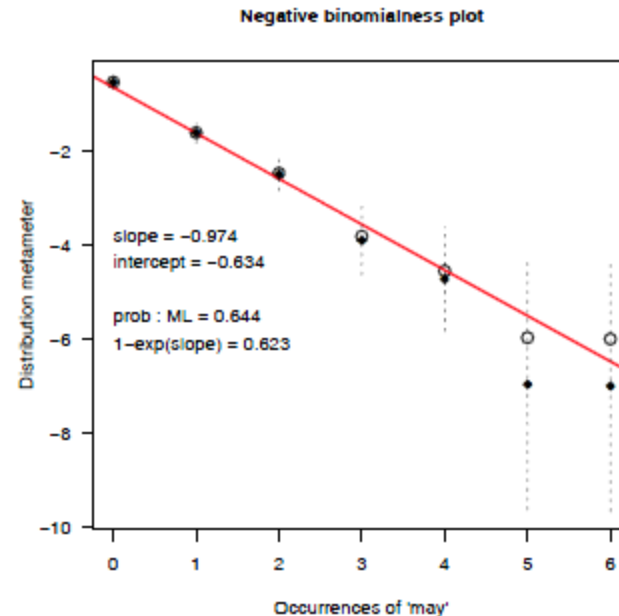
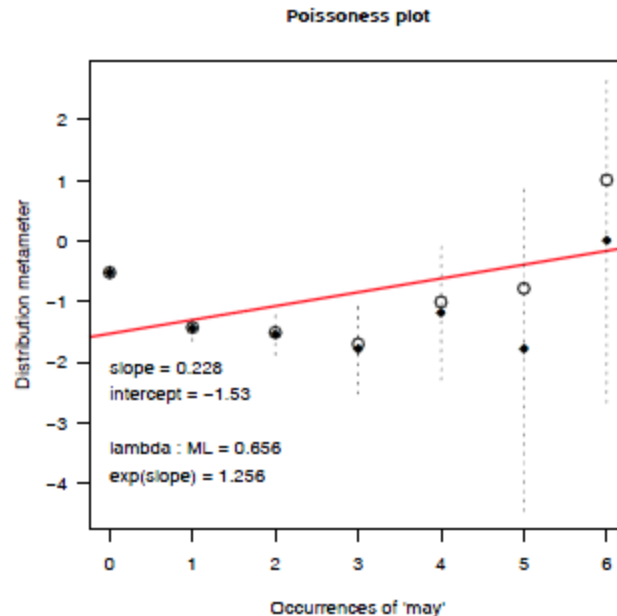
Source: adapted from Hoaglin and Tukey (1985), Table 9-15.

- In these plots, a straight line confirms that the data follow the given distribution.
- Confidence intervals around the points indicate **uncertainty** for the count metameter.
- The slope and intercept of the line give **estimates** of the distribution parameters.

distplot: Federalist

Try both Poisson & Negative binomial

```
distplot(Federalist, type="poisson", xlab="Occurrences of 'may'")  
distplot(Federalist, type="nbinomial", xlab="Occurrences of 'may'")
```

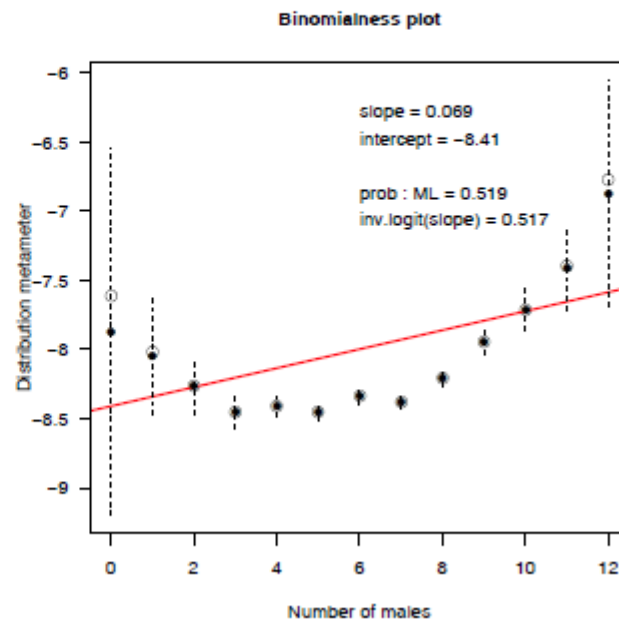
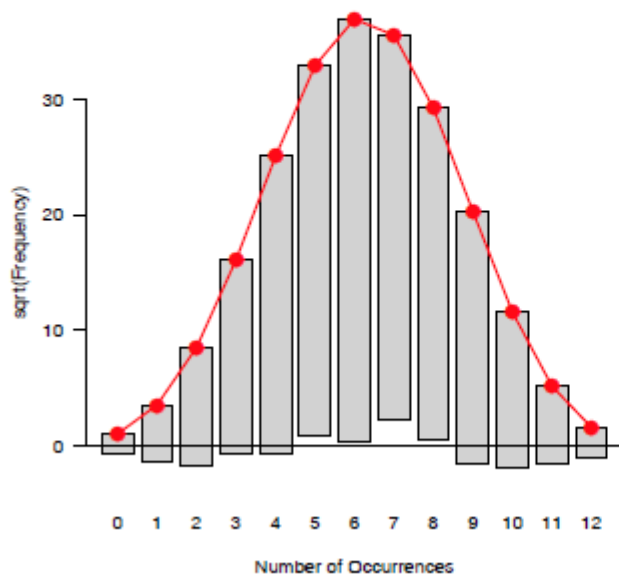


Again, the Poisson distribution is seen not to fit, while the Negative binomial appears reasonable.

distplot: Saxony

For purported binomial distributions, the result is a “binomialness” plot

```
plot(goodfit(Saxony, type="binomial", par=list(size=12)))  
distplot(Saxony, type="binomial", size=12, xlab="Number of males")
```



Both plots show heavier tails than the binomial distribution. `distplot()` is more sensitive in diagnosing this

What have we learned?

Main points:

- Discrete distributions involve basic *counts* of occurrences of some event occurring with varying *frequency*.
- The ideas and methods for one-way tables are building blocks for analysis of more complex data.
- Commonly used discrete distributions include the binomial, Poisson, negative binomial, and logarithmic series distributions, all members of a *power series* family.
- Fitting observed data to a distribution $\rightarrow$ fitted frequencies, $N\hat{p}_k$, $\rightarrow$ goodness-of-fit tests (Pearson X^2 , LR G^2)
- R: `goodfit()` provides `print()`, `summary()` and `plot()` methods.
- Plotting with rootograms, Ord plots and generalized distribution plots can reveal *how* or *where* a distribution does not fit.

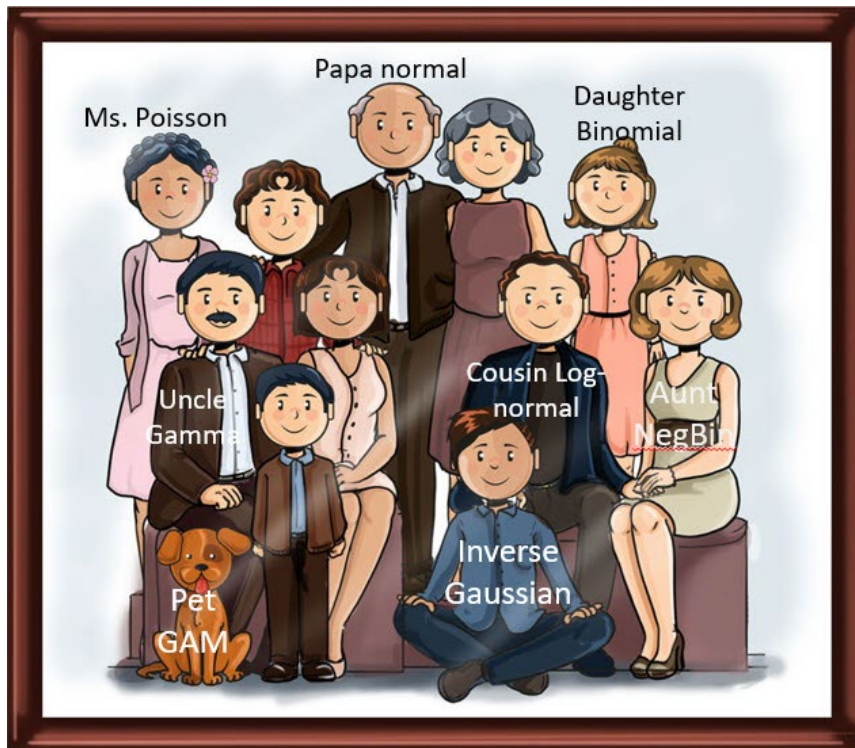
What have we learned?

Some explanations:

- The Saxony data were part of a much larger data set from Geissler (1889) (Geissler in [vcdExtra](#)).
 - For the binomial, with families of size $n = 12$, our analyses give $\hat{p} = \Pr(\text{male}) = 0.52$.
 - Other analyses (using more complex models) conclude that p varies among families with the same size.
 - One explanation is that family decisions to have another child are influenced by the boy–girl ratio in earlier children.
- As suggested earlier, the lack of fit of the Poisson distribution for words in the Federalist papers can be explained by [context](#) of the writing:
 - Given “marker” words appear more or less often over time and subject than predicted by constant rates (λ) for a given author (Madison or Hamilton)
 - The negative binomial distribution fit much better.
 - The estimated parameters for these texts allowed assigning all 12 disputed papers to Madison.

Looking ahead: The GLM family

These discrete distributions are also part of a larger family, the **exponential family**
This is the basis for the extension to generalized linear models, the **GLM family**



Families share:

- **Same structure:** canonical form, “natural” parameters
- **Sufficient statistics:** all you need to estimate
- **Conjugate priors** in Bayesian analysis

Families allow:

- Generic **estimation**, like `glm()`
- Common **diagnostics** and **graphical** methods

Families provide: ❤️ mutual love , 👨‍👩‍👧 sense of belonging & identity, procreation!

Looking ahead: PhdPubs data

Example 3.24 in DDAR gives data on the number of publications by PhD candidates in the last 3 years of study

```
data("PhdPubs", package = "vcdExtra")  
table(PhdPubs$articles)
```

```
##  
##      0      1      2      3      4      5      6      7      8      9     10     11     12     16     19  
## 275 246 178  84  67  27  17  12   1   2   1   1   2   1   1
```

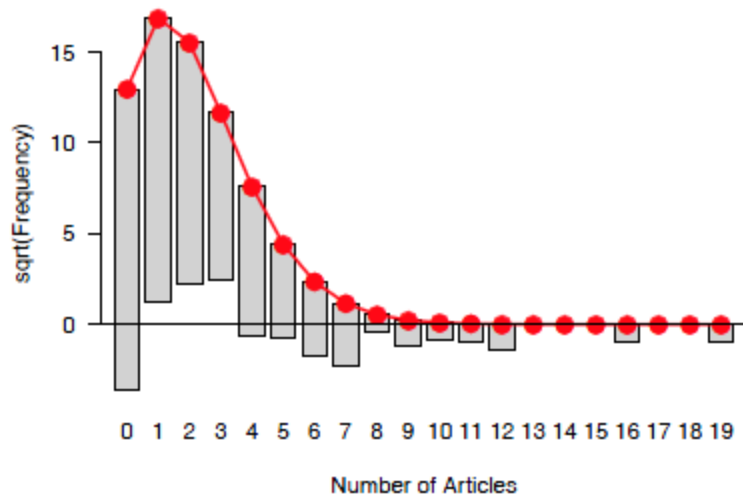
N = 915, mean(articles) = 1.69

- There are **predictors**: gender, marital status, number of children, prestige of dept., # pubs by student's mentor
- We fit such models with **glm()**, but need to specify the **form** of the distribution
- Ignoring the predictors for now, a baseline model could be
`glm(articles ~ 1, data=PhdPubs, family = "poisson")`

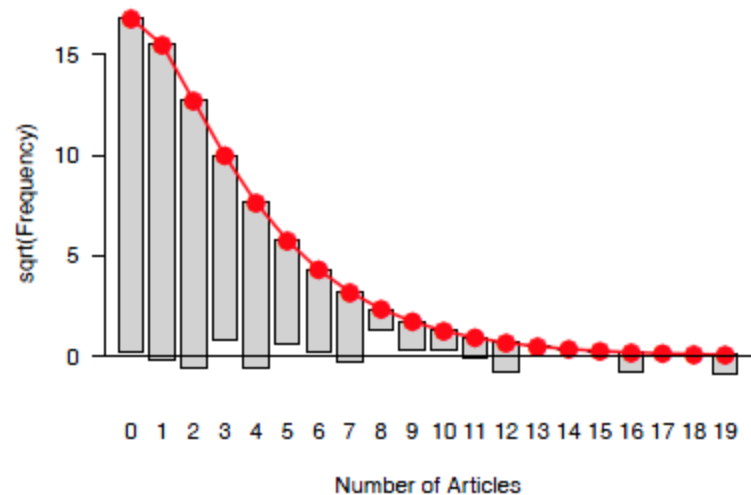
Looking ahead: PhdPubs

```
plot(goodfit(PhdPubs$articles), xlab = "Number of Articles",  
     main = "Poisson")  
plot(goodfit(PhdPubs$articles, type = "nbinomial"),  
     xlab = "Number of Articles", main = "Negative binomial")
```

Poisson



Negative binomial



Poisson doesn't fit: Need to account for **excess 0s** (some never published)
Neg binomial: Sort of OK, but should take predictors into account

Looking ahead: Count data models

Count data regression models (DDAR Ch 11)

- Include predictors
- Allow different distributions for **unexplained** variation
- Provide tests of one model vs. another
- Special models handle the problems of **excess zeros**: `zeroInfl()`, `hurdle()`

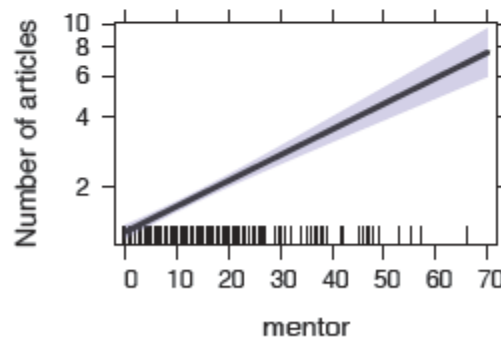
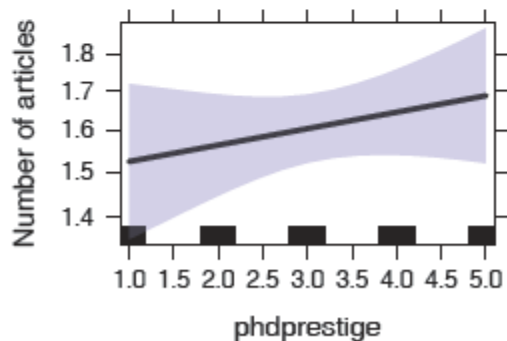
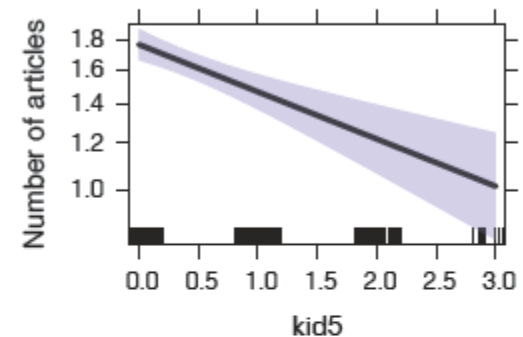
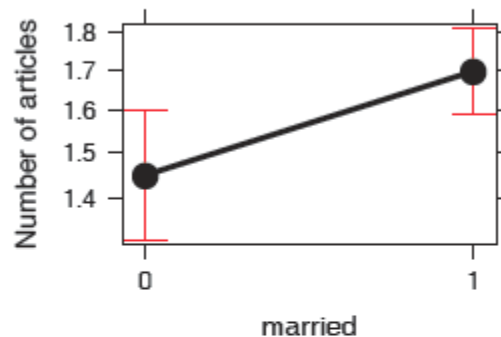
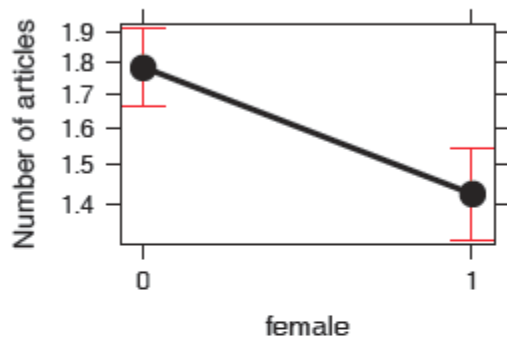
```
# predictors: female, married, kid5, phdprestige, mentor
phd.pois <- glm(articles ~ ., data=PhdPubs, family=poisson)
phd.nbin <- glm.nb(articles ~ ., data=PhdPubs)

LRstats(phd.pois, phd.nbin)

## Likelihood summary table:
##           AIC   BIC LR Chisq  Df Pr(>Chisq)
## phd.pois 3313 3342    1634  909    <2e-16 ***
## phd.nbin 3135 3169    1004  909    0.015 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


Looking ahead: Effect plots

Effect plots show the predicted values for each term in a model, averaging over all other factors.



These are better visual summaries for a model than a table of coefficients.

Summary

- Discrete distributions are the building blocks for categorical data analysis
 - Typically consist of basic counts of occurrences, with varying frequencies
 - Most common: binomial, Poisson, negative binomial
 - Others: geometric, log-series
- Fit with `goodfit()`; plot with `rootogram()`
 - Diagnostic plots: `Ord_plot()`, `distplot()`
- Models with predictors
 - Binomial → logistic regression
 - Poisson → poisson regression; logliner models
 - These are special cases of **generalized** linear models